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Flexural behavior of reinforced high strength concrete slabs containing glass powder, flay ash, and rice husk ash as cement substitute

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ABSTRACT

This study supports a move toward greener construction methods by endorsing sustainable building materials, which can have long-term positive effects on the environment and public health. Utilizing waste materials makes concrete production more environmentally friendly by lowering emissions, landfill usage, and the need for natural resources. The purpose of this study is to evaluate the effect of using glass powder (GP), fly ash (FA), and rice husk ash (RHA) as cement substitutes on the flexural behavior of reinforced high-strength concrete (RHSC) slabs. Ten RHSC slabs with different proportions (0 %, 10 %, 15 %, and 20 %) of GP, FA, and RHA as cement substitutes were experimentally cast and tested under four-line loading conditions. Experimental results indicated that using FA and RHA as cement alternatives for HSC mixes reduced workability, whereas GP increased it. Test results showed that the HSC mixes containing 15 % GP, 15 % FA, and 10 % RHA had the highest increase in compressive, tensile, and flexural strengths. Incorporating 15 % GP, FA, or RHA enhanced the ultimate load, stiffness, and toughness of the tested RHSC slabs compared to the control slab. Conversely, increasing replacement levels of GP, FA, or RHA to 20 % of cement led to a reduction in ultimate load. The optimal replacement ratios for the RHSC slab are 15 % GP, 15 % FA, or 10 % RHA, which result in acceptable performance of RHSC slabs and protect the environment and natural resources. Finally, the predictions of two methodologies, ACI 318 and ECP 203, were compared with the experimental findings, revealing satisfactory concordance.

1. Introduction

The most common material is concrete, and at least 8 % of the world's CO_2 emissions come from the cement sector [1,2]. The increase in social and economic activities has been closely linked to the growth of the construction industry, which has led to an increase in demand for Portland cement. Therefore, the development of eco-friendly materials has garnered increasing attention to mitigate the environmental impact of the cement manufacturing process. This, in turn, is essential for reducing global emissions and producing environmentally friendly concrete, which are two of the industry's main goals [3–6]. The use of waste materials as partial substitutes for cement in concrete production provides an environmentally viable remedy to the challenges linked to cement manufacturing. These materials offer the possibility of economical, high-performance concrete while promoting the circular economy through the recycling of industrial and agricultural waste. Research and development progress will likely expand their application, fostering advancements in sustainable construction methodologies [7, 8].

Over the past few years, researchers have conducted several studies to assess the impact of using industrial and agricultural waste materials such as glass powder (GP), fly ash (FA), and rice husk ash (RHA) as cement substitutes. The authors [9–12] stated that the inclusion of GP as a substitute for cement improves the workability of concrete. The particle size of GP markedly affects the workability of concrete mixtures [13–16]. The authors [17–22] determined that increased GP levels correlate with enhanced compressive strength, reaching a maximum of 25 % GP as a cement alternative. The enhancement in compressive strength was contingent upon the particle size of GP. The authors [23, 24] indicated that mixes containing GP frequently demonstrated superior flexural and tensile strengths compared to the control mixture. The studies [25–27] indicated that an increase in GP content correlates with a decrease in the percentage of water absorption.

The global concrete industry has effectively utilized FA, primarily as a mineral additive in Portland cement concrete and as a blended cement component [28,29]. The authors [30–32] concluded that the

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incorporation of FA as a cement substitute enhances workability. Yurdakul et al. [33] discovered that adding more class F fly ash to binary mixes made them easier to work with. They think this is because the Class F fly ash made the particles less likely to rub against each other. According to the researchers [33–36], the compressive, tensile, and flexural strengths of concrete significantly improved with the addition of optimum levels (15 %-25 %) of FA. The improvement level depended on the fineness of the FA; the higher the fineness, the greater the improvement in concrete strengths. Higher fineness may enhance pozzolanic activity and concrete density, improving mechanical properties. Nayak et al. [31] found that using 25 %–30 % FA improved durability and resistance to environmental factors such as freezing and thawing. Bahedh and Jaafar [37] reported that the permeability, workability, and compressive strength of ultra-high-performance concrete increased by increasing the FA level as a cement replacement. Nath and Sarker [38] observed that the strength of high-strength concrete (HSC) continued to improve significantly, and the FA concretes exhibited superior resistance to chloride ion penetration.

One of the waste products from the agriculture sector is RHA, which may be recycled into valuable building materials and reduce the energy needed to create cement. The authors [39–41] noted that RHA adversely affected the workability of concrete mixes, indicating a higher water content in concrete containing RHA compared to those made exclusively with Portland cement. The main features of RHA particles that increase water demand in concrete are their round shape and fine size [42]. The trend showed that as the RHA level increased, the workability of concrete mixes decreased linearly, attributed to the higher surface area of RHA grains [43,44]. RHA often improves the mechanical properties of concrete, demonstrating a consistent increase in strength up to a certain concentration, beyond which it begins to decline. The authors [45–48] stated that the ideal RHA level is 10 %, whereas Sathe et al. [49] propose an optimum of 15 %. According to references [50–54], the tiny RHA particles contribute to the density of the microstructure and reduce acids, sulfates, chlorides, alkali-silica reactions, and carbonation, thereby enhancing the concrete's strength. Amin et al. [45] reported that a 10 % replacement ratio of RHA achieved strong bond strength between steel and concrete.

Several studies investigated the behavior of reinforced concrete structures that incorporated waste materials as cement substitutes. In their study [55], Elsayed et al. found that adding GP up to 10 % greatly improved the axial capacity and toughness of the tested columns. In another study, Elsayed et al. [56] found that incorporating 10 % GP as a replacement for cement and 5 % crumb rubber significantly enhanced the capacity of unheated columns. Raising the GP level to 20 % exerted a slight impact on the column capacity. The increased temperatures adversely affect the column capacity. Hama et al. [57] reported that the incorporation of 15 % GP as a partial substitute for cement significantly enhanced the flexural capacity of the examined beams. Elsayed et al. [58] pointed out that the incorporation of waste aluminum fibers improved the flexural and shear capacities of recycled coarse aggregate concrete beams incorporating GP. Mustafa et al. [59] demonstrated that incorporating 10 % GP increased the slab's capacity by 8.8 %, whereas the addition of 20 % GP resulted in a reduction of ultimate load by 14.7 %. Yassen et al. [60] found that incorporating GP enhanced the strength capacity of the beam, as indicated by the improved ultimate load of beams with GP relative to the reference beam.

Alghazali et al. [61] conducted an experimental investigation into the shear strength of full-scale self-compacting concrete (SCC) beams containing a high volume of FA. Their findings showed a slight difference in the shear strength of SCC beams as cement substitute ratios increased. According to Elsayed et al. [62], the SCC mixes can benefit from a combination of ternary binders consisting of 50 % cement, 25 % FA, and 25 % slag. The high level of cement replacement has a minimal impact on shear strength while enhancing the ductility of the SCC beam. Hashmi et al. [63] concluded that the flexural performance of high-volume fly ash concrete (HVFAC) structural beams and slabs improves as concrete strength increases. RC beams and slabs have demonstrated good serviceability, with up to 40 % FA replacement in higher-grade concrete mixes. Gollakota et al. [64] emphasized that FA-modified concrete has superior resistance to sulfate attacks and alkali-silica reactions, making it a viable alternative for aggressive environments. Yoo et al. [65] pointed out that 35 % fly ash replacement achieved the best flexural performance of RC beams, balancing structural efficiency and environmental benefits. In their investigation into the shear strength of full-scale beams made from high-volume FA concrete (HVFAC), Arezoumandi and Volz [66] found that the HVFAC beam containing 70 % Class C fly ash outperforms the control beam in terms of shear strength. Khaliq and Kodur [67] examined the performance of HSC columns with FA exposed to elevated temperatures. The results of the fire resistance tests revealed that the fire resistance of FA concrete columns is nearly identical to that of conventional HSC columns.

Ahsan et al. [68] confirmed that finer RHA demonstrated greater concrete strength compared to coarse RHA. Additionally, the 10 % RHA replacement was preferable to 20 % in concrete properties. The authors [69,70] examined the shear response of RHA in RC beams, demonstrating an increase in strength with rising RHA content up to a specific threshold, beyond which deterioration occurs. Both studies determined that a 10 % RHA ratio is optimal for shear strength.

Although previous studies have investigated the impact of GP, FA, and RHA as cement replacements in various concrete applications, comprehensive studies examining their effects on reinforced high-strength concrete (RHSC) slabs are lacking. Prior studies indicate that substituting GP, FA, and RHA for cement enhances concrete strength; however, most of these investigations have primarily focused on compressive strength, workability, and durability. Their influence on flexural performance, stiffness, and toughness in RHSC slabs remains unclear. This research aims to bridge this gap by evaluating the effects of different GP, FA, and RHA proportions as cement substitutes in RHSC slabs. The findings are expected to expand the existing knowledge base and provide practical insights into the sustainable utilization of these materials in high-performance concrete applications.

2. Research significance

In the last few years, the production of environmentally friendly concrete has become a key issue for protecting the environment from carbon dioxide emissions and preserving natural resources. Numerous studies have examined the mechanical properties of concrete that incorporate GP, FA, or RHA. However, there is still little research that aims to understand the performance of concrete elements, including GP, FA, and RHA. To fill this gap, this research focuses on the flexural behavior of RHSC slabs that incorporate varying amounts of GP, FA, and RHA as a cement substitute.

3. Experimental work

3.1. Material

Ordinary Portland cement (OPC) (CEM I 42.5 N) that complied with ASTM C150 [71] was utilized. Its specific surface area is 332 m^2/Kg, its specific gravity is 3.15, and its initial setting time is 170 minutes, with a final setting time of 241 minutes. Fig. 1 presents the use of GP, FA, and RHA as cement substitutes. The specific gravity and specific surface area of GP are 2.95 and 324 m^2/Kg, respectively. FA is a fine, powdery material with a specific gravity of 2.31 and a specific surface area of 393 m^2/Kg. RHA was partially used as a cement substitute; it has a specific gravity of 2.12 and a specific surface area of 283 m^2/Kg. Husks undergo a burning process at approximately 700°C to manufacture RHA. Table 1 shows the chemical composition of OPC, GP, FA, and RHA. The coarse aggregate used was crushed stone. It was cleaned with potable water to make sure the coarse aggregates remained dust-free. As per ASTM C127 [72], the properties of the coarse aggregate were evaluated,

Fig. 1. GP, FA, and RHA.

Table 1
The chemical composition of OPC, GP, FA, and RHA.

Chemical Oxides (%)	SiO_2	CaO	Fe_2O_3	SO_3	Al_2O_3	MgO	K_2O	Na2O	P_2O_5	CL	TiO_2	LOI	Cr2o3
OPC	20.38	62.44	3.75	2.53	4.77	1.25	0.23	0.43	-	0.04	-	3.43	0.02
GP	72.4	11.50	0.065	0.20	1.43	0.32	0.35	13.65	-	-	0.035	-	0.002
FA	76.35	0.55	3.69	0.12	14.72	0.54	0.95	0.20	0.15	0.012	0.62	0.52	-
RHA	77.82	11.78	1.35	1.26	1.09	0.70	1.94	0.04	1.14	0.45	0.17	5.07	-

and it was sieved to obtain aggregates of different sizes, as illustrated in Table 2. Natural sand was used as the fine aggregate. Table 2 presents the evaluation of the fine aggregate's properties by ASTM C128 [73]. Fresh tap water, free from impurities, was used. A superplasticizer was added to maintain the necessary level of workability in the concrete mixes and has been developed mainly for use in ready-mixed and precast concrete applications where the greatest levels of performance and durability are needed. Silica fume, a very fine pozzolanic material, is produced as a by-product of silicon metal or ferrosilicon alloy production. It is commonly used in several cementitious products, such as mortar, shotcrete, grout, and concrete. The experimental work utilized steel reinforcement with a 10 mm diameter for longitudinal reinforcement. According to ASTM A370 [74], the properties of the steel were determined, and the results are presented in Table 3.

3.2. Concrete mix design

Ten concrete mix proportions were designed by the American Concrete Institute (ACI 318–19) [75], as illustrated in Table 4. One concrete mix without a substitute was kept as a control mix, while nine concrete mixes with three replacement levels of 10 %, 15 %, and 20 % of GP, FA, and RHA as cement substitutes were designed. To evaluate the compressive strength and tensile strength, three cubes measuring 150 × 150 × 150 mm were cast, along with three cylinders with a diameter of 100 mm and a height of 200 mm. A beam without reinforcement, measuring 500 mm in length, 100 mm in width, and 100 mm in height, was cast to measure the flexural strength of each concrete mix.

3.3. Test specimens

Ten RHSC slabs (one slab for each HSC mix) were cast and tested. The slabs had dimensions of 1000 mm in length (with an effective span of 850 mm), 500 mm in width, and 100 mm in thickness. Each slab maintains a tension reinforcement ratio of 0.59 % or 10 mm at 150 mm. Moreover, each slab maintains a secondary bottom reinforcement ratio of approximately 10 mm at 190 mm. The thickness of the concrete cover for the steel reinforcement was 25 mm. Fig. 2 details the specifics, measurements, and reinforcement of the slabs.

Table 2
Physical Properties of the used aggregates.

Property	Coarse aggregate	Fine aggregate
Grading (mm)	4.75–22.1	0.16–2.35
Specific Gravity	2.70	2.60
Bulk Density (t/m^3)	1.60	1.65
Fineness modulus%	0.30	0.90
Water absorption%	1.35	1.70

Table 3
Mechanical properties of reinforcing steel bars.

Diameter (mm)	Actual Area (mm^2)	Yield Strength (MPa)	Yield strain	Ultimate Strength (MPa)	Young's modulus (GPa)
10	78.1	430	0.00248	642.8	205

3.4. Test Setup

Fig. 3 describes the test setup for the specimens. The slab surfaces were coated in white paint before testing to enhance the visibility of cracks. A hydraulic jack of 900 kN, with 0.25 mm displacement control, was used to apply the load vertically to the slab. With an effective span of 850 mm, the simply supported RC slabs were tested under four-line loads until they failed after 28 days of casting. Three linear variable differential transducers (LVDTs) were placed on the bottom sides of the slabs to record the deflection.

4. Result and discussion

4.1. Mechanical properties of concrete

4.1.1. Workability

A slump test for HSC mixes was conducted according to ASTM C143 [76], as listed in Table 5. The results indicate that the inclusion of GP and FA as substitutes for cement enhanced the workability of HSC mixes; in contrast, the addition of RHA adversely affected their workability. Using 10 %, 15 %, and 20 % of GP resulted in increases in slump values of 12.8 %, 21.1 %, and 30.6 %, respectively, in comparison to the HSC control mix. The experimental results align with similar observations made by authors [9,10,16]. It may be attributed to the fact that GP has a crystalline structure and doesn't absorb water. Additionally, GP possesses superior surface properties and improves particle size dispersion, suggesting that incorporating more GP into the formulation could account for the observed enhancement in workability [16]. The

Table 4
Mix proportions of concrete (kg/m^3).

Mix code	Cement	GP	FA	RHA	Silica Fume	coarse aggregate	Fine aggregate	Water	superplasticizer
*M1 (Control)	450	0	0	0	45	1085	650	170	8
M2(10 %GP)	405	45	0	0	45	1085	650	170	8
M3(15 %GP)	382.5	67.5	0	0	45	1085	650	170	8
M4(20 %GP)	360	90	0	0	45	1085	650	170	8
M5(10 %FA)	405	0	45	0	45	1085	650	170	8
M6(15 %FA)	382.5	0	67.5	0	45	1085	650	170	8
M7(20 %FA)	360	0	90	0	45	1085	650	170	8
M8(10 %RHA)	405	0	0	45	45	1085	650	170	8
M9 (15 %RHA)	382.5	0	0	67.5	45	1085	650	170	8
M10(20 %RHA)	360	0	0	90	45	1085	650	170	8

* Control Mix

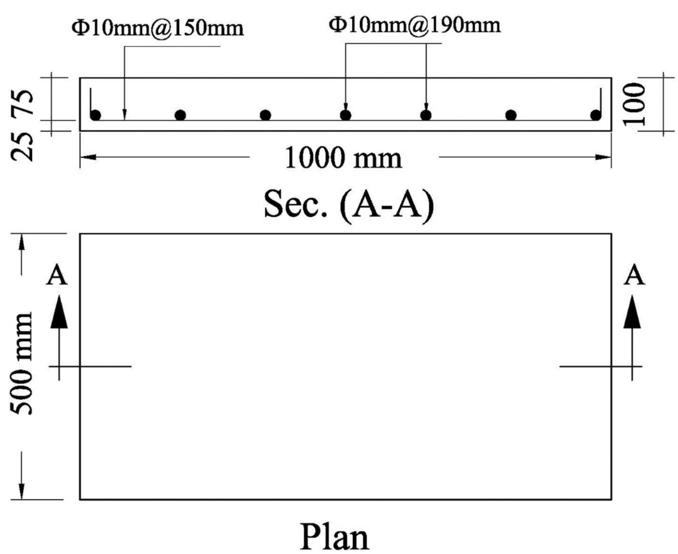

Fig. 2. Details and dimensions of the tested RC slabs, (all dimensions in mm).

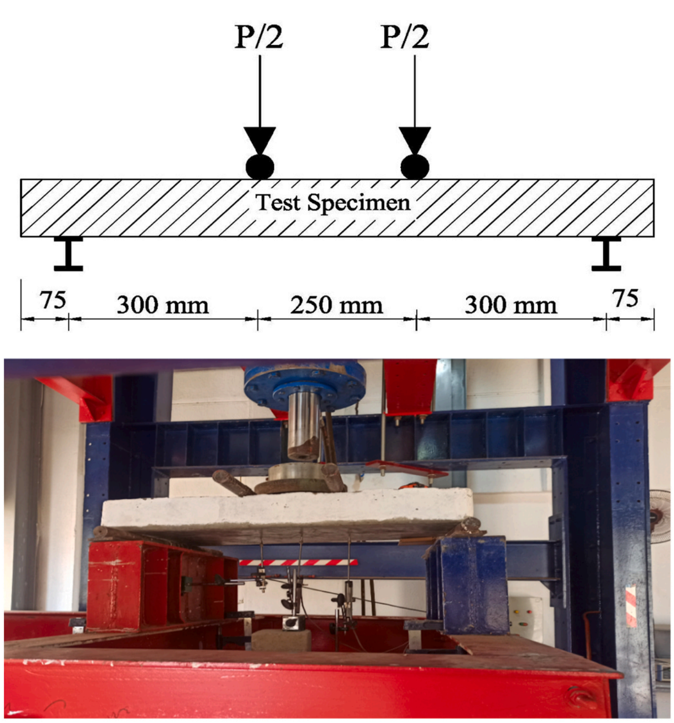

Fig. 3. Test setup of the tested slabs.

incorporation of 10 %, 15 %, and 20 % of FA reduced the workability of the HSC mixes; compared to the HSC control mix, the slump values increased by 8.9 %, 16.7 %, and 24.4 %, respectively. It may be ascribed to FA particles being often spherical or glassy, which reduces friction between cement particles [31,32]. The small, smooth particles of FA exhibit lower water absorption than cement particles. This promotes improved flow and mixing, hence enhancing the overall workability of the concrete. The author's observations [30,31,38] confirm these findings. The addition of RHA to HSC mixes resulted in a reduction in their workability. The slump values went down by 14.4 %, 21.7 %, and 31.1 % compared to the HSC control mix when 10 %, 15 %, and 20 % of RHA were added, respectively. RHA absorbs water quickly, and measurements of its particles' surface area show that they are smaller than cement's. The experimental results agree with the author's findings [39, 69].

4.1.2. Compressive strength

According to EN 12390–3 [77], Table 5 lists the compressive strength of various HSC mixes. In comparison to the HSC control mix, the compressive strength increased by 5.1 % and 8.6 % by using 10 % and 15 % GP, respectively. This phenomenon is attributed to the filler effect of GP, which facilitates the micro-filling of voids between concrete components. This enhances the density of the concrete and increases its compressive strength [15,22]. While 20 % GP showed a slight decrease of 1.3 %. The results agree with the conclusions drawn in the references [9,10]. When cement replaced 10 % and 15 % of the FA, the compressive strengths increased by 1.0 % and 8.6 %, respectively; however, the percentage decreased by 6.4 % when 20 % FA was present. It may be due to FA being widely recognized for its pozzolanic properties, which contribute to the long-term enhancement of concrete performance by refining the microstructure and increasing the formation of calcium silicate hydrate (C-S-H) gel [36]. The results agree with the conclusions mentioned in the references [41]. The compressive strength was increased by 8.9 %, 8.1 %, and 3.2 % by using 10 %, 15 %, and 20 % of RHA, respectively. The findings align with the studies published in [45–47]. It can be explained by the pozzolanic properties of RHA reacting with cement hydrates to create calcium silicate hydrates (C-S-H). This is a substance that improves the bonds and cohesiveness between molecules by micro-filling voids. This makes concrete denser by filling in the gaps between its parts [41,48]. It was observed that the 15 % FA achieved slightly higher resistance than the 10 % RHA, followed by the 15 % GP.

4.1.3. Splitting tensile strength

Table 5 shows the splitting tensile strength of various HSC mixes, which was conducted by C496 [78]. After 28 days, the tensile strength increased by 8.9 % and 13.3 %, respectively, when 10 % and 15 % GP were used, compared to the HSC control mix. However, a slight decrease of 4.4 % was observed when 20 % GP was used. The enhancement in tensile strength can be attributed to the incorporation of GP, which refines the microstructure of concrete, leading to a denser interfacial

Table 5
Fresh and Hardened properties of HSC mixes.

Mix #		*M1	M2	M3	M4	M5	M6	M7	M8	M9	M10
Slump (mm)		180.0	203.0	218.0	235.0	196.0	210.0	225.0	154.0	141.0	124.0
Cube Compressive Strength (fcu) (MPa)	Individual	62.3	64.3	69.0	62.5	64.3	67.9	56.0	70.6	67.8	66.0
		62.2	65.6	67.3	62.0	62.1	68.0	60.3	66.2	67.7	64.0
		63.5	68.1	68.0	61.2	63.4	68.4	58.9	68.0	68.0	64.1
	Average	62.7	66.0	68.1	61.9	63.3	68.1	58.4	68.3	67.8	64.7
	SD	0.6	1.6	0.7	0.5	0.9	0.2	1.8	1.8	0.1	0.9
	COV	0.9 %	2.4 %	1.0 %	0.9 %	1.4 %	0.3 %	3.1 %	2.6 %	0.2 %	1.4 %
Splitting Tensile Strength(ft) (MPa)	Individual	4.5	4.0	4.9	4.0	3.6	5.6	4.2	5.2	4.3	5.1
		4.8	5.0	4.3	5.1	4.2	4.1	5.0	4.6	4.7	4.2
		4.1	5.7	6.2	3.9	5.7	5.1	4.1	5.4	4.8	3.9
	Average	4.5	4.9	5.1	4.3	4.5	4.9	4.4	5.1	4.6	4.4
	SD	0.3	0.7	0.8	0.5	0.9	0.6	0.4	0.3	0.2	0.5
	COV	6.4 %	14.2 %	15.4 %	12.5 %	19.6 %	12.6 %	9.1 %	6.7 %	4.7 %	11.6 %
Flexural Strength (MPa)	Individual	5.5	5.3	7.0	4.6	6.0	7.5	4.3	7.6	5.0	4.3
		5.0	6.7	6.2	5.6	6.9	6.6	4.9	7.1	6.1	5.1
		5.9	6.6	9.3	5.7	5.9	6.9	4.1	8.3	6	5.7
	Average	5.5	6.2	7.5	5.3	6.3	7.0	4.4	7.7	5.7	5.0
	SD	0.4	0.6	1.3	0.5	0.4	0.4	0.3	0.5	0.5	0.6
	COV	6.7 %	10.3 %	17.5 %	9.4 %	7.2 %	5.3 %	7.7 %	6.4 %	8.7 %	11.4 %

transition zone, which contributes to improved tensile properties [20,21,24]. The tensile strength increased by 0.4 % and 8.9 %, respectively, when cement was substituted for 10 % and 15 % of the FA. The enhancement in tensile strength can be attributed to the pozzolanic activity of FA, which improves molecular bonding and cohesion within the cement matrix [36]. On the other hand, when 20 % FA was added, the percentage decreased by 2.2 %. By applying 10 % and 15 % of RHA, respectively, the tensile strength increased by 13.3 % and 2.2 %. It may be due to RHA significantly improving the transition zone between aggregates and the cementitious matrix by reducing microcracks, which facilitates better adhesion and tensile resistance [41,48]. The tensile strength decreased by 2.2 % upon the addition of 20 % RHA. These findings confirm that the optimum cement replacement percentages for improved tensile performance are 15 % GP, 15 % FA, and 10 % RHA, as observed in this study.

4.1.4. Flexural strength

The flexural strength of HSC mixes was conducted according to ASTM C78 [79], as listed in Table 5. The flexural strength increased by 12.7 % and 32.7 % compared to the HSC control mix when 10 % and 15 % GP were used, respectively. It can be explained by adding GP, which improves concrete's pore structure, creating a denser, more compact microstructure that greatly increases flexural resistance [13,20]. However, it went down by 3.6 % when 20 % GP was used. When 10 % and 15 % FA were substituted for flexural strengths, the increase was 14.5 % and 27.3 %, and the addition of 20 % FA had a 20 % decrease. The reference [36] highlighted that FA particles reduce porosity and increase overall strength by filling gaps in the hydration products. Flexural strengths decreased by 8.1 % in the mix with 20 % RHA and increased by 40 % and 3.6 % in the cases of 10 % and 15 % RHA. It may be due to RHA ash increasing adhesion in the interfacial transition zone, strengthening the link between aggregates and the cement matrix [41,48]. These findings validate that optimal replacement percentages for flexural strength enhancement are 15 % GP, 15 % FA, and 10 % RHA, as observed in this study.

4.2. Results of tests on reinforced concrete slabs

4.2.1. Failure modes and cracking pattern

Fig. 4 displays the failure modes and crack patterns observed in each tested RHSC slab. Cracks formed on the tension side of the tested slabs. The cracks began as a linear formation in the transverse direction, originating at the middle of the slab and beneath the line loads. As the applied load increases, the crack rises, and new cracks begin to appear and spread. In the mid-span, increasing the load also resulted in more flexural cracks. The last loading stage caused pre-existing cracks to widen, some slabs to have a falling cover, and a considerable decrease in the rate at which new cracks developed. Table 6 displays that the addition of GP delayed the initiation of cracks and controlled their progression. The first crack load was 60 % of the failure load for the RHSC control slab (S1). It was 61 %, 68 %, and 66 % for RHSC slabs (S2, S3, and S4), which have 10 %, 15 %, and 20 % GP, respectively. Mustafa et al. [59] reported a similar observation as the addition of GP had a beneficial effect on the first cracking load. FA was used to delay the appearance of cracks in the RHSC slabs. RHSC slabs (S5, S6, and S7), which have different amounts of FA (10 %, 15 %, and 20 %), made up 49 %, 62 %, and 56 % of the failure load, in that order. Adding 20 % FA had a beneficial effect on the first cracking load. It was found that RHSC slabs (S8, S9, and S10) with 10 %, 15 %, and 20 % RHA each had a failure load of 69 %, 56 %, and 53 %, respectively. The addition of RHA with 10 % had a beneficial effect on the first cracking load. 20 % RHA (S10) had higher failure strength than 10 % GP (S2) and 20 % FA (S7). When GP, FA, and RHA are added, the compressive and tensile strengths go up. This has a direct effect on how well the slab can stop cracks from starting and spreading.

4.2.2. Load-deflection curves

Fig. 5 illustrates the relationships between load and deflection (mid-span) for RHSC slabs. The load-deflection curves showed three stages; it is evident that the slabs in the first stage, referred to as the elastic stage, behaved linearly from the initial loading up to the first crack load. After the slabs reached their initial crack load in the first stage, the load-deflection curve became nonlinear because of the cracking in the second stage. All the slabs reached their maximum load capacity before they completely collapsed, leading to a decrease in bending strength as the displacement increased. The deflection at failure in RHSC slabs S2, S3, and S4 was 6.1 %, 2.6 %, and 6.6 % less than in RHSC control slab S1. This was because they had different amounts of GP (10 %, 15 %, and 20 %). At the same time, the deflection at failure went down by 3.5 %, 0.7 %, and 11.0 % in RHSC slabs S5, S6, and S7, which have 10 %, 15 %, and 20 % FA, respectively. When 10 %, 15 %, or 20 % RHA was used instead of cement, RHSC slabs S8, S9, and S10 had 5.8 %, 12.4 %, and 13 % less deflection at failure, respectively. The ultimate deflection of RHSC slabs decrease when GP, FA, and RHA are used as cement substitutes, as shown in Fig. 6. This reduction in the ultimate deflection is related to the enhancement of mechanical properties of HSC mixes, which directly influence the slab's stiffness and deformation capacity.

4.2.3. Ultimate Load

Table 6 presents the ultimate load of the tested slabs. The

Fig. 4. Failure modes and cracking pattern in tested slabs at failure.

incorporation of GP, FA, and RHA up to 15 % led to an enhancement in the ultimate load of RHSC slabs. This improvement may be attributed to a denser microstructure and increased concrete strengths, which are essential for the load-bearing capacity of concrete elements. The ultimate load of RHSC slabs S2 and S3, which incorporated 10 % and 15 % of GP, increased by 2.2 % and 4.3 %, respectively, in comparison to the control slab S1. The ultimate load of the RHSC slabs (S5 and S6) with 10 % and 15 % FA increased by 4.8 % and 6.3 %, respectively, compared to the control slab (S1). The addition of 10 % and 15 % RHA to RHSC slabs (S8 and S9) resulted in an increase in the ultimate load by 6.3 % and 2.5 %, respectively, when compared to the control slab (S1). The introduction of 20 % GP, FA, or RHA resulted in a decrease in the ultimate load of the RHSC slab. The ultimate loads of RHSC slabs (S4, S7, and S10), which incorporated 20 % GP, FA, and RHA, were reduced compared to the control slab (S1) by approximately 11.3 %, 7.1 %, and 2.1 %, respectively.

4.2.4. *Stiffness*

The initial stiffness of the tested slabs was computed from the load-deflection curves, as listed in Table 6. Incorporating GP, FA, and RHA as cement substitutes enhanced the stiffness of RHSC slabs. The pozzolanic reaction of GP, FA, and RHA particles filling up capillary pores could explain this. This could make RHSC slabs stiffer. This process likely results in a denser microstructure, improving mechanical properties,

Fig. 4. (*continued*).

especially the increase in compressive strength and modulus of elasticity, which are crucial components of structural rigidity and responsible for the observed improvement in stiffness. The RHSC slabs containing 15 % GP, 15 % FA, and 10 % RHA demonstrated the highest stiffness levels. The stiffness of the RHSC slabs (S2, S3, and S4), containing 10 %, 15 %, and 20 % GP improved by 13.9 %, 16.4 %, and 3.0 %, respectively, compared to the control slab (S1). Compared to the control slab (S1), the stiffness of RHSC slabs (S5, S6, and S7) containing 10 %, 15 %, and 20 % FA was improved by about 16.4 %, 26.4 %, and 23.9 %, respectively. It was found that the stiffness of RHSC slabs (S8, S9, and S10) with 10 %, 15 %, and 20 % of RHA was 34.3 %, 23.9 %, and 7.5 % higher than that of the control slab (S1), respectively.

4.2.5. *Toughness*

Toughness is a good indicator of how ductile the slabs are; it is the specimen's capacity to endure deformations up to the failure load. Toughness is measured as the area under the load-deflection curve up to the failure load, as listed in Table 6. In comparison to the control slab (S1), a slight enhancement in the toughness of RHSC slabs (S2 and S3) of approximately 1.8 % and 3.6 % was noted with the introduction of GP at 10 % and 15 %, respectively; however, a reduction of approximately 18.2 % occurred with the addition of 20 % GP. When 10 % and 15 % FA

Table 6
Experimental results of the tested beams.

Slab No.	Mix code	First cracking load (kN)	Ultimate load (kN)	Ultimate deflection (mm)	Initial Stiffness (kN/mm)	Toughness (kN.mm)
S1	M1*Control	40.0	67.2	31.5	6.7	1656.2
S2	M2 (10 %GP)	41.8	68.7	29.75	7.6	1685.0
S3	M3 (15 %GP)	47.9	70.1	30.75	7.8	1715.5
S4	M4 (20 %GP)	39.2	59.7	29.50	6.9	1354.5
S5	M5 (10 %FA)	34.8	70.4	30.50	7.8	1718.1
S6	M6 (15 %FA)	44.4	71.7	32.0	8.5	1790.5
S7	M7 (20 %FA)	35.0	62.5	28.0	8.3	1451.6
S8	M8 (10 %RHA)	49.0	71.4	30.0	9.0	1750.6
S9	M9 (15 %RHA)	38.3	68.9	27.75	8.3	1494.7
S10	M10 (20 %RHA)	34.8	65.8	27.50	7.2	1391.1

were added, the toughness of RHSC slabs (S4 and S5) went up by 3.8 % and 8.1 %, respectively, compared to the control slab (S1). However, when the FA levels went up to 20 %, the toughness went down by 12.4 %. Substituting 10 % RHA of cement for RHSC slab S8 resulted in a 5.7 % increase in toughness compared to the HSC control slab S1. On the other hand, the toughness of S9 and S10 with 15 % and 20 % RHA decreased by 9.8 % and 16.0 %, respectively. The results show that replacement ratios of 15 % GP, 15 % FA, and 10 % RHA cement in RHSC slabs significantly increased its toughness. It may be due to the combined pozzolanic reactions enhance the concrete's capacity to resist cracking and improve energy absorption after cracking, which is a key component of toughness.

5. Codes equations for RC slabs containing waste materials

This section is intended to estimate the flexural capacity of the RHSC slabs utilizing the American building code ACI 318–19 [75] and the Egyptian code of practice (ECP 203–18) [80]. The aim of this point is to evaluate the alignment between the design code formulas and the results of the experimental study.

5.1. American building code ACI 318–19 [75]

(As shown in Fig. 7)
Take $\varepsilon_{cr} = 0.1\varepsilon_0$ assume $\varepsilon_0 = 0.003 \therefore \varepsilon_{cr} = 0.0003$

$$C_C = T_s + T_1 + T_2 \tag{1}$$

$$C_C = 0.85 * f_c' * a * b \tag{2}$$

$$T_s = A_s * f_y \tag{3}$$

$$T_1 = 0.5 * f_t * x * b \tag{4}$$

$$T_2 = 0.5 * f_t * y * b \tag{5}$$

$$a = \beta_1 * c \tag{6}$$

$$\beta_1 = 0.85 - 0.00005(f_c' - 4000) \tag{7}$$

$$x = \frac{\varepsilon cr}{\varepsilon s}(d - c) \tag{8}$$

$$y = 10\frac{\varepsilon cr}{\varepsilon s}(d - c) \tag{9}$$

$$y_1 = c - 0.5a + \frac{2}{3}x \tag{10}$$

$$y_2 = c – 0.5a + x + \frac{y}{3} \tag{11}$$

$$M_{uinternal} = Ts\ (d - 0.5\ a) + T1 * y1 + T2 * y2 \tag{12}$$

$$M_{u\ internal} = M_{u\ external} \tag{13}$$

$$Pt = \frac{2\ \ Muexternal}{ashear\ span} \tag{14}$$

5.2. Egyptian code of practice (ECP 203) [80]

(As shown in Fig. 8)
Take $\varepsilon_{cr} = 0.1\varepsilon_0$ assume $\varepsilon_0 = 0.003 \therefore \varepsilon_{cr} = 0.0003$

$$C_C = T_s + T_1 + T_2 \tag{15}$$

$$C_C = 0.67 * f_{cu} * a * b \tag{16}$$

$$T_s = A_s * f_y \tag{17}$$

$$T_1 = 0.5 * f_t * x * b \tag{18}$$

$$T_2 = 0.5 * f_t * y * b \tag{19}$$

$$x = \frac{\varepsilon cr}{\varepsilon s}(d - c) \tag{20}$$

$$y = 10\frac{\varepsilon cr}{\varepsilon s}(d - c) \tag{21}$$

$$y_1 = c - 0.5a + \frac{2}{3}x \tag{22}$$

$$y_2 = c – 0.5a + x + \frac{y}{3} \tag{23}$$

$$M_{uinternal} = Ts\ (d - 0.5\ a) + T1 * y1 + T2 * y2 \tag{24}$$

$$M_{u\ internal} = M_{u\ external} \tag{25}$$

$$Pt = \frac{2\ \ Muexternal}{ashear\ span} \tag{26}$$

Where:

Cc: Compression force in the concrete compression zone	T_s: Tensile force in tension reinforcement
T_1 and T_2: Tensile force in the concrete	f_t: Concrete tensile strength
ε_c: Ultimate concrete strain	ε_s: Maximum steel strain
ε_{cr}: Cracking strain	f_y: Yield stress of main steel
b: Beam breadth	d: Effective depth
fc′: Cylinder compressive strengths (MPa)	fcu: Cube compressive strengths (MPa)
a: Equivalent stress block depth	$\beta 1$: Stress block factor
M_u: Ultimate moment	$M_{u\ external}$: External moment
$M_{u\ internal}$: Internal moment	Pt: total load

5.3. Comparison between experimental and analytical results

Table 7 displays the experiment results and the analytical values predicted by ACI 318–19 [75] and ECP 203–20 [80]. The results show a significant consistency in the ultimate load calculations between the ACI 318–19 [75] and ECP 203–20 [80]. The results indicated that the

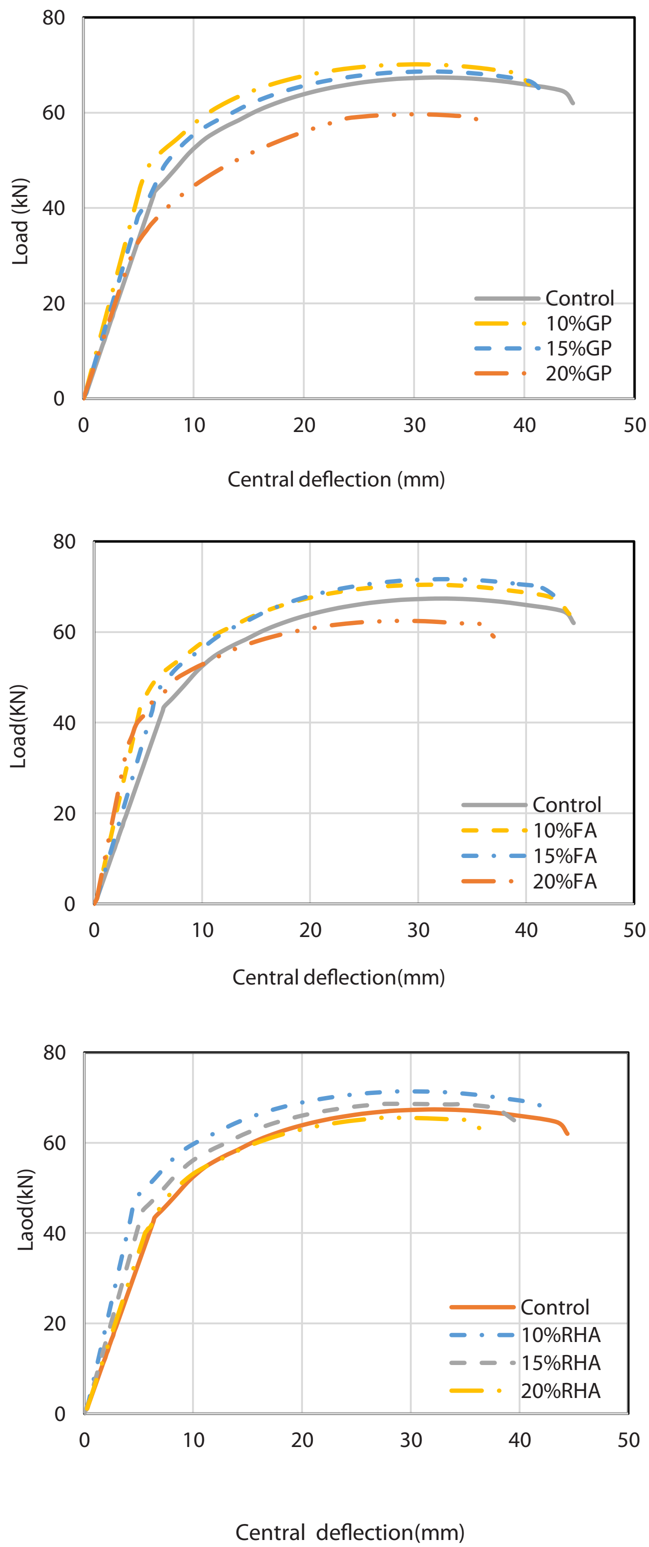

Fig. 5. Load–central deflection curves of tested slabs.

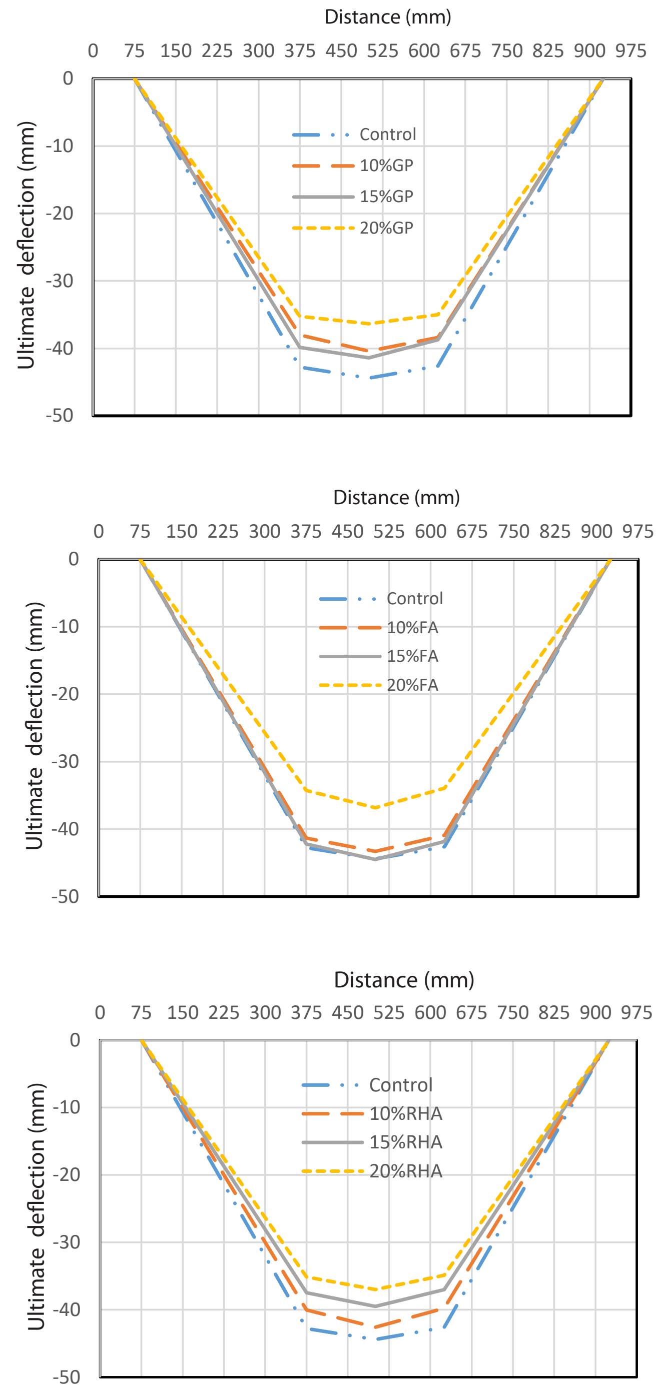

Fig. 6. Deflection along the slabs at ultimate load.

ultimate load of the tested slabs from both design codes was mostly similar. The average, standard deviation, and coefficient of variation of the ratio between the experimental ultimate load and the analytical one are 1.03, 0.05, and 5.3 %, respectively. This suggested that both codes slightly underestimate the ultimate load of the slabs, but the predictions are relatively close. The design of RHSC slabs with GP, FA, and RHA is similar to traditional slabs. The overall flexural behavior of HSC with GP, FA, and RHA slabs is similar to those of traditional slabs when it comes to how loads deflect, cracks spread, and the way they break.

6. Conclusions

In this study, the experimental results for ten RHSC slabs that contain GP, FA, or RHA under flexure tests are examined, the following conclusions are made in light of the experimental results:

1. Incorporating GP and FA as cement replacements improved the workability and consistency of HSC mixes; conversely, adding RHA had a detrimental impact on their workability.

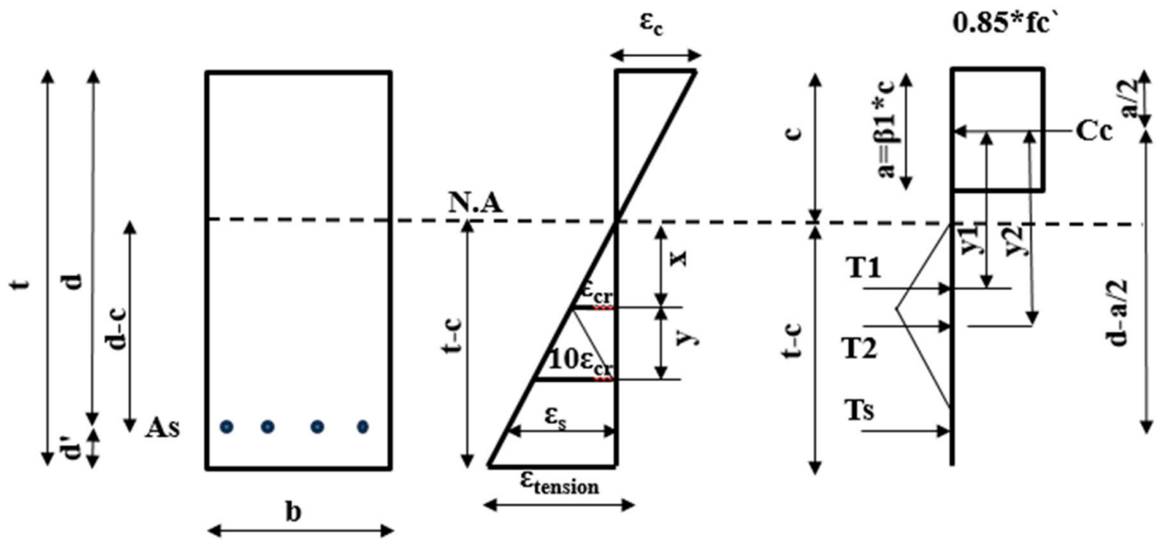

Fig. 7. The American Building Code specifies net tensile strain and strain distribution.

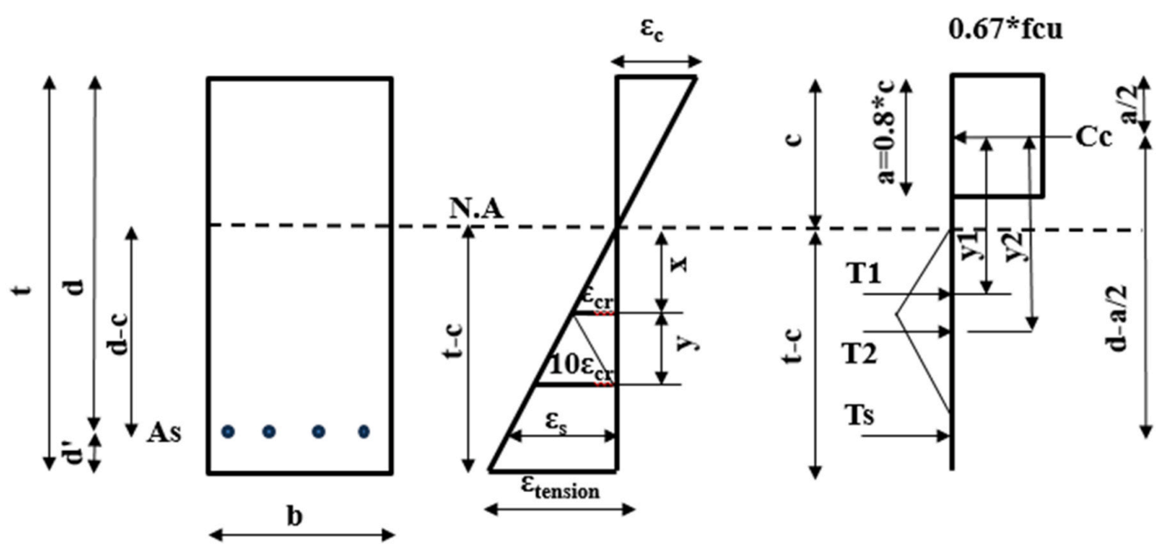

Fig. 8. The Egyptian code specifies the strain distribution and net tensile strain.

Table 7
Comparison between experimental and analytical results.

Slab No.	Ultimate load (kN)			Exp./ACI	Exp./ECP.
	Exp.	ACI 318–19	ECP.203		
S1	67.2	65.5	65.0	1.03	1.03
S2	68.7	66.2	65.7	1.04	1.05
S3	70.1	66.6	66.1	1.05	1.06
S4	59.7	65.2	64.7	0.91	0.92
S5	70.4	65.5	65.1	1.07	1.08
S6	71.7	66.3	65.9	1.08	1.09
S7	62.5	65.1	64.7	0.96	0.97
S8	71.4	66.5	66.1	1.07	1.08
S9	68.9	65.9	65.4	1.04	1.05
S10	65.8	65.5	64.9	1.00	1.01
Average				1.03	1.03
Standard deviation (S.D)				0.05	0.05
Coefficient of Variation (C.O.V)				5.3 %	5.2 %

2. Incorporating GP, FA, and RHA up to 15 % as cement replacement improved the compressive, tensile, and flexural strengths of HSC mixes. However, adding 20 % of GP, FA, and RHA slightly reduced HSC strengths.
3. The ultimate load and toughness of RHA slabs demonstrated an enhancement as the replacement percentage of GP, FA, and RHA rose by 15 %. On the other hand, raising the replacement levels to 20 % resulted in a slight reduction in ultimate load and toughness.
4. The research on HSC mix substitution ratios shows that RHSC behaviors get better and sustainability goals are met. Incorporating 15 % GP, FA, or RHA enables the RHSC slabs to attain satisfactory ultimate load, stiffness, and toughness.
5. Incorporating GP, FA, and RHA enhanced the stiffness of RHSC slabs. The RHSC slabs containing 15 % GP, 15 % FA, and 10 % RHA demonstrated the highest stiffness.
6. The results of the experiments and the predictions from the ACI 318–19 and ECP 203–20 code equations were relatively close to each other, with an average, standard deviation, and coefficient of variation of 1.03, 0.05, and 5.3 %, respectively.

7. The results show that GP, FA, and RHA can improve the performance of reinforced concrete slabs, such as in precast panels and structural walls. GP, FA, and RHA are recommended for marine and humid environments, as it has been proven to reduce permeability, enhance durability and improve resistance to chemical attacks.

7. Future Research Recommendations

The authors suggest recommendations and directions for future research:

- Studying the impact of GP, FA, and RHA on fire resistance and thermal insulation could open new possibilities for sustainable construction.
- Conducting full-scale structural elements to assess the real-world performance of GP, FA, and RHA-modified concrete.
- Exploring the combined use of GP, FA, and RHA with fiber reinforcement (e.g., steel, glass, basalt, and carbon fibers) to enhance crack resistance and flexural toughness.

CRediT authorship contribution statement

Saad Mustafa: Writing – original draft, Investigation, Formal analysis, Data curation. **Elsayed Mahmoud:** Writing – review & editing, Supervision, Methodology. **Hassanean Mahmoud M.:** Writing – review & editing, Supervision, Methodology.

Declaration of Competing Interest

We pledge that we are not involved in the interests of the financial, commercial, legal, or professional relationship with other organizations, or with the people we worked with them, that could influence my research

Data Availability

Data will be made available on request.

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